SM2501/FSG3084 Biomechanics of human movement, 7.5 hp 2025

Welcome to the course in Biomechanics of Human Movement at KTH! This document describes the course aims, teaching activities and the basis for grading. The examination will consist of a series of homework projects and a final group project (written reports and oral presentations). Note: the PhD student course SG3084 is given in parallel to this course with one additional project.

What is Biomechanics of human movement?

Biomechanics is a large, and ever-growing field that broadly can be defined as applying the use of mechanics to study living systems. Many different definitions have been proposed, covering topics as wide as fluid flow, tissue engineering, joint modeling, *in vivo* measurements, etc. For this course, we can apply Hay's definition (1973):

> *Biomechanics is the science that examines forces acting upon and within a biological structure and effects produced by such forces*

wherein forces include internal forces or external forces, which must be measured or modeled accurately, and may be attributed to movements of segments or tissue deformation. The study of biomechanics of human movement may include the mechanics that produce movement, analysis of movement and forces, prediction of movement, etc.

This course

This course will focus on movement of the human body and accompanying forces, fundamentals of motion production, analysis, and modeling. The course focuses first on the anatomy and physiology of the human muscle-force system, then on biomechanical applications involved in human movement. From this foundation, the course will focus on methods to analyze biomechanical problems, including kinematics and kinetics of movement and the muscle force system. Relevant topics requiring such analyses will be discussed. Emphasis is placed not only on computation, but also on interpretation and critical evaluation of results.

Objectives

This course is aimed towards upper level undergraduates and masters students.

After the course, the student should be able to:

- Apply principles of classical mechanics to the study of human motion
- Describe motion with precise, well-defined mechanical and anatomical terminology
- Describe the internal and external forces acting on the body during typical human activities
- Understand how muscle actions control movements
- Understand the dynamic relationship between structure and function in the musculoskeletal system
- Model muscle activation and human movement
- Understand the methods and limitations of different experimental and analytical techniques used

Instruction

Instruction will be in the form of 2-3 lectures per week (a total of 17 lectures), at least one 1-hour session of experiments in the Promobilia MoveAbility Lab, and ~3 computer labs (2 hours each).

Lectures will be held in lecture rooms.

Computer labs will be hybrid. You can participate from wherever you like - you are welcome to participate in the computer room if you prefer. *Bring headphones with a microphone.*

TA Office Hours: Jenny Maier (post doc, TA) will hold office hours for help with OpenSim projects, at KTH MoveAbility as per the schedule, or at other agreed-upon dates and times. She would appreciate if you contact her ahead of time at jenmai@kth.se so she knows who to expect.

Prerequisites

Students are expected to have complete courses in basic mechanics and dynamics or have equivalent knowledge corresponding to an undergraduate degree in engineering. No previous coursework in anatomy or physiology is required.

Literature

The course book will be *Biomechanics of Movement; The Science of Sports, Robotics, and Rehabilitation,* by Thomas K. Uchida and Scott L. Delp. ISBN 9780262044202. **It's available online at the KTH Library**, also from internet bookstores, both as a book and as an e-book (though the e-book is more expensive). It's also possible to rent the e-book via the publisher (MIT Press, see https://mitpress.mit.edu/books/biomechanics-movement)

While this book is quite good, it doesn't cover quite all of the lecture material. I'm including a list of other recommended literature, which can either be borrowed from me for short periods or borrowed from libraries. Hamill, Knutzen and Derrick is an excellent overall introduction to biomechanics and functional anatomy, and was the previous course book. Robertson et al. is good for experimentalists and researchers, but has very little introduction to the area. Some other complementary reading assignments may be suggested.

- Biomechanical Basis of Human Movement, 4th Ed, J. Hamill, K.M. Knutzen and T. Derrick.
- Research Methods in Biomechanics, by D.G.E Robertson, G.E. Caldwell, J Hamill, G. Kamen, and S.N. Whittlesey
- Biomechanics and Motor Control of Human Movement, 4th Ed, by David Winter
- Kirtley: Clinical Gait Analysis – Theory and Practice
- Nordin & Frankel: Basic Biomechanics of the Musculoskeletal System, 3rd Ed

Software

The open source program OpenSim version 4.5 from SimTK will be used in several assignments. It is a free, open-source software (https://simtk.org/projects/opensim -> downloads). It is installed in the computer labs, and I recommend you install it on your own computers.

Course Evaluations

At the end of the course, a written evaluation will be distributed.

Course Staff

The course is given by faculty at KTH MoveAbility Lab, Dept. of Engineering Mechanics, with guest lecturers from Karolinska Institutet/Danderyd Hospital and from the Swedish School of Sport and Health Sciences (GIH).

Main instructor:	Professor Lanie Gutierrez-Farewik	lanie@kth.se	790 7719
Assistant teacher:	Assoc. Professor Ruoli Wang	ruoli@kth.se	790 6801
TA:	Dr. Jennifer Maier, post doc	jenmai@kth.se	

Contact

The course page is on the KTH Canvas course web.

Schedule

We have 2-3 lecture times per week, 3 computer labs, 1 data collection at the Promobilia MoveAbility Lab.

Examinations

There are a total of 4 project assignments - 3 project/'homework' assignments during Period 3 and one final project, all to be performed in a group of 2-3 students. You may form your own groups of 3 initially, and I will assign any remaining students to groups. Participation in the data collection at the KTH MoveAbility Lab is compulsory. All members of each group will get the same grade for that project.

Projects/assignments. Information and data for all assignments will be uploaded throughout the period. The assignments vary in how demanding they are, but they are all rather demanding. Each assignment has a due date approximately two weeks after the previous and *all are compulsory to get a final grade*. I expect this due date to be fulfilled unless we have made a previous agreement. Exceptions will only be considered for unavoidable travel, illness, etc.

All projects are *compulsory*. The first and third projects are to be submitted in *written* form, and the second is to be *presented, recorded and sent to the teacher* – refer to each project description for specific details. Emphasis is placed on synthesis and interpretation of results, not simply on the computations. Programming code is to be attached as an appendix when relevant.

Final project: The project will also be completed in groups and presented orally only. Groups will be assigned later in the period.

About group work: The concept with group work is that the *group* works together on the project, not that each person does 1/3 of the project. That wouldn't be group work. I know that students usually distribute the work at some point so that one person can take more of a lead on part of the analyses or figures or text, but each person should still understand the entire project and **be able to answer questions about any part**. Also, you usually have to do the first part to be able to do the second, etc., so dividing up the tasks will make it practically impossible for someone who hasn't been involved in part 1 to write anything sensible about part 2, and the whole

group's grade will likely suffer. Students in the same group may have different backgrounds, but this should be used as a strength - you can all bring different experience to the group and learn much more when you work *together*.

In short, **for group work, think *multiplication*, not *division***.

In cases of severe group dysfunction, students should contact the instructor. Peer evaluations may be used to adjust individual grades in exceptional circumstances.

Grading (SM2501): A grade of between 0-10 points will be given for each of projects 1-3, depending on the ambition achieved and the assignment quality. All members in a group will be given the same grade for that project. The final project will be graded between 0-20 points. This makes a total of 50 points of which 20 are required to pass the course. Furthermore, a minimum of 2 points must be earned on each homework assignment to pass the course. Grading:

20-24	25-30	31-36	37-42	43-50
E	D	C	B	A

Grading (FSG3084 for PhD students): Only Pass/Fail grades are possible. To achieve a Pass, students must complete one extra project individually, and achieve an overall equivalent grade of A or B as per the points above.

Academic integrity

- **No use of generative AI tools allowed.** Any use of generative AI tools is prohibited in the course project reports and presentations, as generative AI use will conflict with the learning objectives and graded examinations. Students are advised to avoid generative AI even when studying.
- All submitted work must be original and created by the group
- Proper citations required for all sources
- Code sharing between groups is not permitted. You may discuss code with other groups but not share.
- Plagiarism will result in failure of the assignment and possible course failure. See KTH's information: https://www.kth.se/en/student/stod/studier/fusk-1.997287